MTH4224 Project 2 Report

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1 Introduction

In baseball, specifically in Major League Baseball (MLB), it is essential to find players of similar skill levels. This can be done by looking at their performance over a year or a career. The number and detail in baseball statistics make it relatively easy to do so. However, finding similar players by performance is less straightforward and requires machine learning to see the clusters of comparable players. Adding to this complexity is the vast number of statistics a single player can have, making it initially difficult to gauge how players perform similarly. Still, it is possible to find similar players through clustering.

Since there are a plethora of different clustering techniques, it would be good to use various clustering algorithms to see if one performs well with the given dataset. As for how to make a player's performance more visually understandable, dimension reduction techniques can help to reshape data into 2D and 3D data, instead of the high dimension in which the data is natively. Like with clustering, there are many different algorithms, so using a variety of them should be beneficial for studying. Furthermore, seeing how different player clusters have similar statistics is essential. This can be done using parallel plotting, which graphs the relative value of a specific feature from the original dataset.

This project uses a combination of data dimension reduction and clustering techniques to find clusters of similar-performing MLB players. This data comes from an open-source dataset that allows users to download high-dimensional data for players based on the year. In finding clusters, each combination of techniques is adjusted to fine-tune the optimal hyperparameters involved in clustering. Finally, parallel plotting is performed to see how important features scale between clusters.

2 Methodology

2.1 Data Sampling and Processing

Two datasets were used in this project: a dataset for pitchers and a dataset for batters in the MLB. While there are some similar features between the two datasets, they can have different values since they have different contexts depending on the player's role. The features involved in these datasets are various statistics gathered throughout a season in a particular year for a player. Some of the statistics are quite simple, such as the number of singles, doubles, triples, etc. that a player was responsible for. Other statistics are a little more complex, such as the percentage of times a pitcher threw a strike and had the batter swing and miss. Some of the more important features for respective player types can be found in Table 1 below. In total, 226 stats are used for pitchers and 142 stats are used for batters.

The source of this data comes from Baseball Savant, a website that provides various visual and numeric statistics for players and teams on the game and season levels. They allow users to download historical data from 1950 to the present year. For the data used in this project, the dataset contained pitchers and batters from 2015 to 2025. This is because more stats are given after 2015, and the number of players shouldn't be too large for either set. The data was downloaded as a csv file, which allowed for a straightforward data importing process into Python.

For the processing of the data, each dataset had any players who had more than 25 percent of their total data missing removed from the dataset. Additionally, if any remaining columns had any missing data, the column was removed. After this, the datasets were normalized to 0 and 1 using `MinMaxScaler` from the

Pitcher Statistics		Batter Statistics	
K%	The percent of batters that have been struck out by the pitcher	AVG	The number of times a batter has resulted in at least a hit single over their total number of times at bat
BB%	The percent of batters that have been walked by the pitcher	xBA	The percent chance that a batter will make any given ball become a hit
Whiff%	The percent at which a pitcher has caused batters to swing in misses over the total times the pitcher has caused batters to swing	OBACON	The percent at which a batter will reach a base after contacting the ball
ERA	The amount of runs the pitcher allows per 9 innings	OBP	The percentage at which a batter has gotten on base over all of their appearances
AVG	The average batting average that the pitcher will make a batter have	K%	The percent at which a batter has struck out over all of their appearances
wOBA	The weighted percent of which batters get on base when playing against the pitcher	xISO	Tries to predict a hitter's power when hitting a given ball
xOBP	The expected percent a batter will get on base when facing the pitcher, depending on the ball	SLG%	The percent of bases a batter has earned over their total appearances
BABIP	The percent of hit balls against the pitcher that don't include home runs or strikeouts	BABIP	The percent of hit balls that don't include home runs or strikeouts
OBS	The on-base percentage plus slugging percentage the pitcher faces	OBS	The on-base percentage plus slugging percentage of the batter

Table 1: Nine of some of the most important statistics that are used to gauge a player's quality

sklearn package. After this preprocessing, the data was reduced to two and three dimensions and then clustered.

3 Dimension Reduction and Clustering

Three algorithms were used to reduce the data's dimensions: PCA, t-SNE, and MDS. These three were chosen to show how the dimensions are reduced in each dataset. Briefly, PCA tries to project the data while maintaining its variance linearly, t-SNE treats the data as non-linear and tries to have similar points stay together as much as possible, and MDS tries to keep the distance between points in the data. There was often different clustering behavior dependent on the reduction method used, since the shape of the outcome data was different between the three.

For the methods used to cluster the data, there were eight algorithms used: Affinity Propagation, Agglomerative Clustering, Birch, DBSCAN, Gaussian Mixture, k-Means, Mean Shift, and Spectral Clustering. These methods contain various treatments of the data in terms of variability, cluster sizes, and linearity. Thus, they provide a good mix of different algorithms. To test the effectiveness of the resulting clusters, three scoring metrics were used: Silhouette score, Davies-Bouldin (DB) score, and Calinski-Harabasz (CH) score. Briefly, the Silhouette score is best for finding the similarity of points in a cluster against other clusters, the DB score measures the similarities between clusters as a whole, and the CH score measures the compactness and separation of clusters.

Furthermore, after the clustering was performed, parallel plots of certain features were made to demonstrate how the clusters were made according to the values of the features. The selected features were seen in Table 1 because they are considered essential in determining a player's quality. The plotting was made by taking the value of each feature analyzed in the graph for each point and the average for the whole cluster.

4 Results

Using the different dimension reduction and clustering techniques, and using the three different scoring metrics to gauge their effectiveness, the resulting plots for pitchers in 2D using PCA can be seen in Figure 1, Figure 2, and Figure 3. For what the same clusters look like in 3D, the plot can be found in Figure 4. Because the 3D clusters usually had worse scores than their 2D counterparts, they weren't used for further analysis. So as not to take up too much of this paper, the other plots of the best scores can be found in the Jupyter notebook given with this paper. A tabulated version of the best scores can be found in Table 2 and Table 3.

The result of the parallel plotting can be found in Figure 5 and Figure 6 for pitchers and batters, respectively. From the pitchers figure, specific patterns of the clusters become more apparent. For one, Clusters 5 and 3 have a high Whiff%, compared to all of the other clusters. Additionally, Clusters 5, 0, and 4 have high amounts of inflicted groundouts, in contrast to Clusters 1 and 2, which have low amounts of inflicted groundouts. The same analysis can be done for the batters parallel plot. For instance, from the plot, we can see that Cluster 4 has a low xISO while Cluster 0 has a high xISO and SLG%.

5 Discussion

5.1 Analysis of Results

The results of the plots are informative about how different reduction and clustering techniques affect the overall clusters of the data. For instance, the DBSCAN algorithm usually has the least number of clusters of all the methods, sometimes even predicting only one cluster. The PCA reduction algorithm shows that it is trying to maintain linearity by placing clusters along two lines. This contrasts t-SNE, which attempts to create well-defined clusters, and MDS, which holds a large spherical shape for the data. In terms of looking at the scores for the clusters, the different datasets give different answers as to which technique is better. For the pitcher dataset, it appears that the t-SNE reduction is best, since it has the highest S-score, lowest DB score, and a relatively high CH score. For the batters, it appears that the PCA technique is best for the same reasons.

Figure 1: The best resulting clusters based on silhouette scores for all clustering algorithms using the PCA reduced pitcher dataset

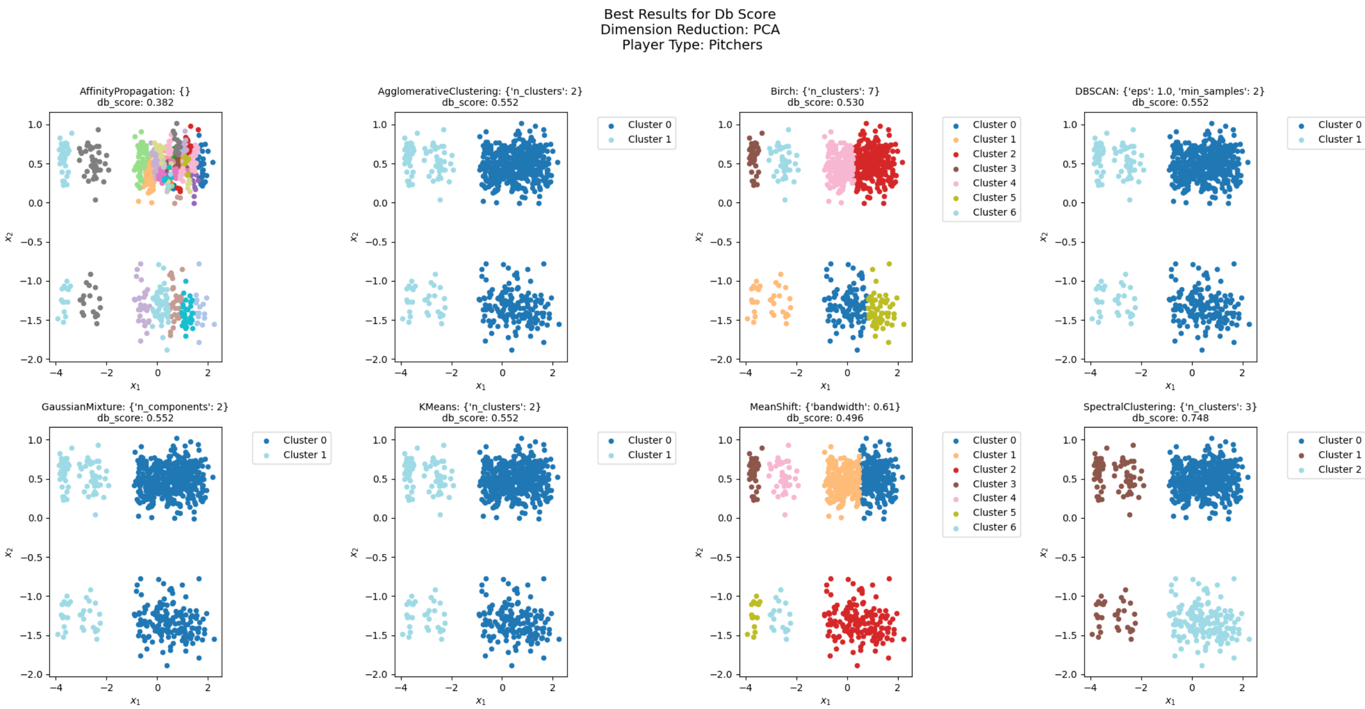

Figure 2: The best resulting clusters based on DB scores for all clustering algorithms using the PCA reduced pitcher dataset

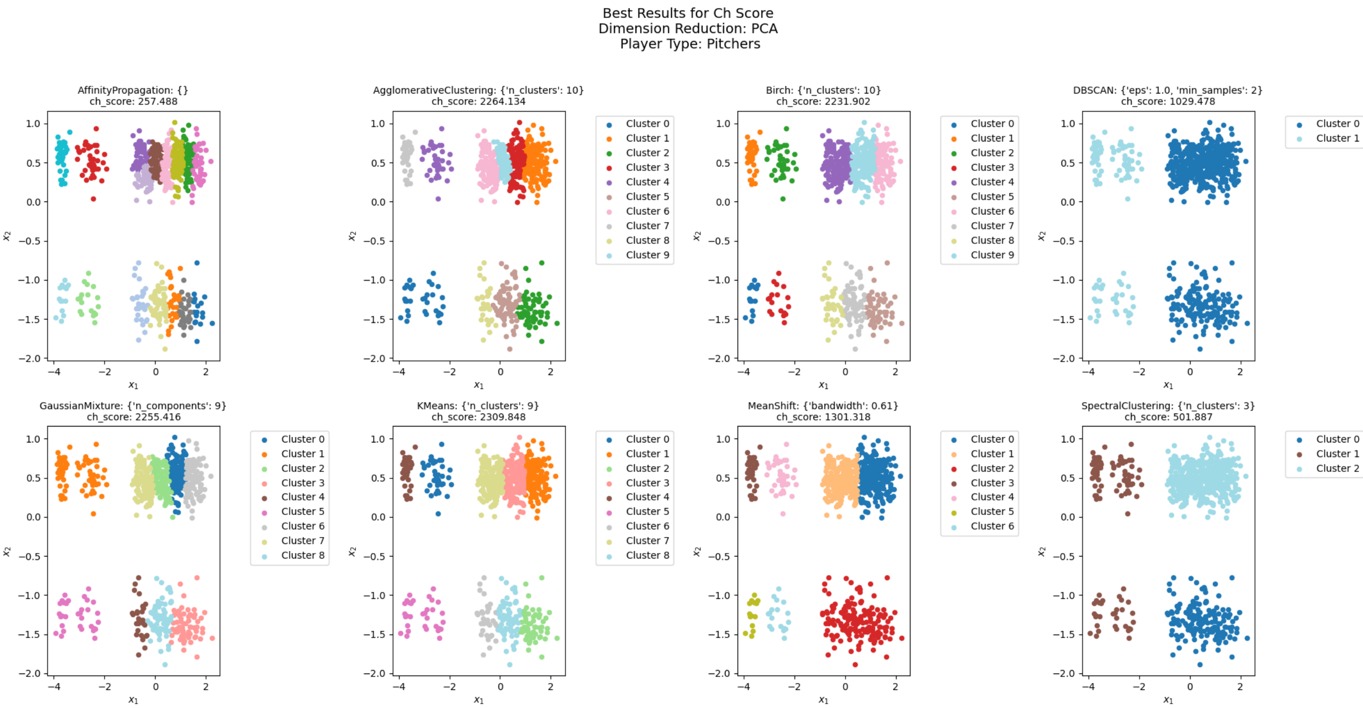

Figure 3: The best resulting clusters based on CH scores for all clustering algorithms using the PCA reduced pitcher dataset

	S-Score (# of Clusters)			DB Score (# of Clusters)			CH Score (# of Clusters)		
	PCA	t-SNE	MDS	PCA	t-SNE	MDS	PCA	t-SNE	MDS
Affinity	0.275(27)	0.377(18)	0.334(23)	0.382(15)	0.824(15)	0.858(23)	257.488(35)	1871.988(16)	720.903(23)
Agglomerative	0.635(2)	0.669(3)	0.436(2)	0.552(2)	0.408(3)	0.755(9)	226.134(10)	1946.615(6)	701.251(9)
Birch	0.635(2)	0.669(3)	0.370(9)	0.530(7)	0.408(3)	0.742(9)	2231.902(10)	1840.184(5)	707.003(9)
DBSCAN	0.635(2)	0.669(3)	-1(1)	0.552(2)	0.408(3)	∞(1)	1029.478(2)	1692.184(3)	-1(1)
Gaussian	0.635(2)	0.669(3)	0.390(9)	0.552(2)	0.408(3)	0.739(9)	2255.416(9)	2050.608(5)	762.250(9)
k-Means	0.635(2)	0.669(3)	0.396(4)	0.552(2)	0.408(3)	0.755(9)	2309.848(9)	2055.406(5)	781.310(9)
MeanShift	0.576(4)	0.669(3)	0.434(2)	0.496(7)	0.408(3)	0.780(2)	1301.318(7)	1916.527(4)	662.502(2)
Spectral	0.534(3)	0.547(2)	0.436(2)	0.802(3)	0.707(5)	0.743(9)	663.797(2)	1669.962(8)	749.703(9)

Table 2: The best of the three different scores based on dim. reduction and clustering technique for pitchers

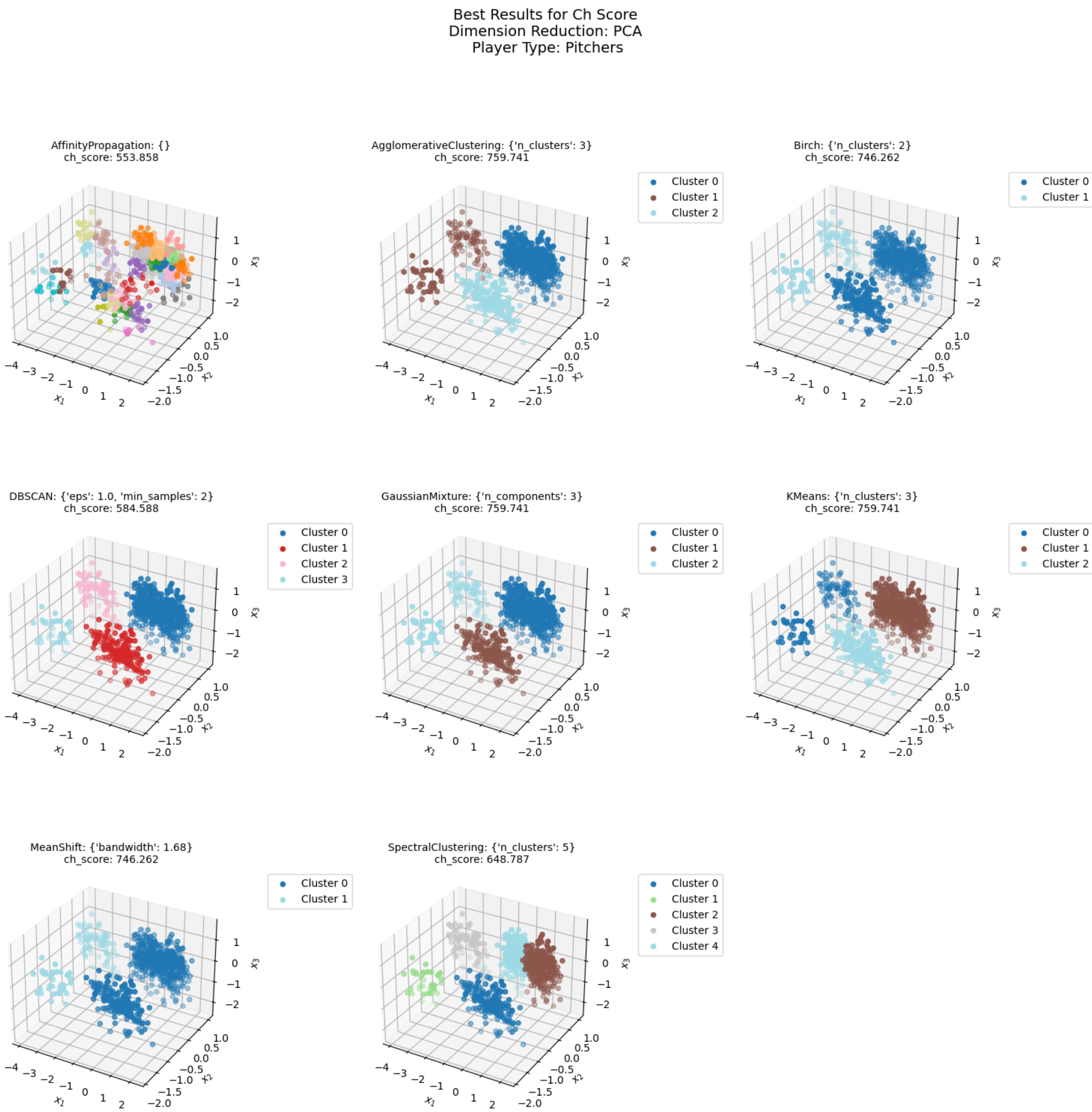

Figure 4: The best resulting clusters based on CH scores for all clustering algorithms using the PCA reduced pitcher dataset

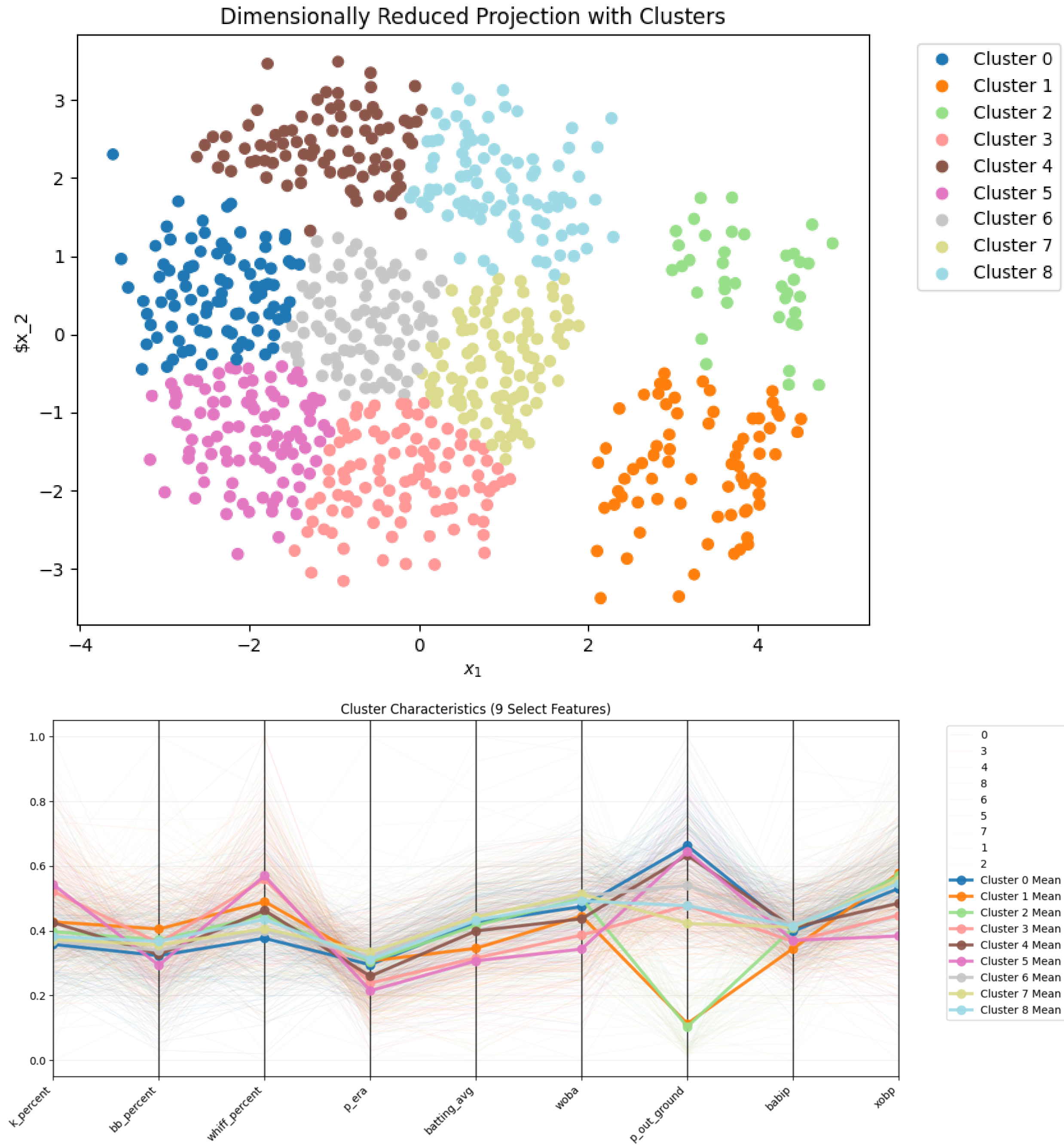

Figure 5: Parallel plot for a cluster resulting from MDS reduction and K-Means clustering. The features chosen are the same features from Table 1 for pitchers.

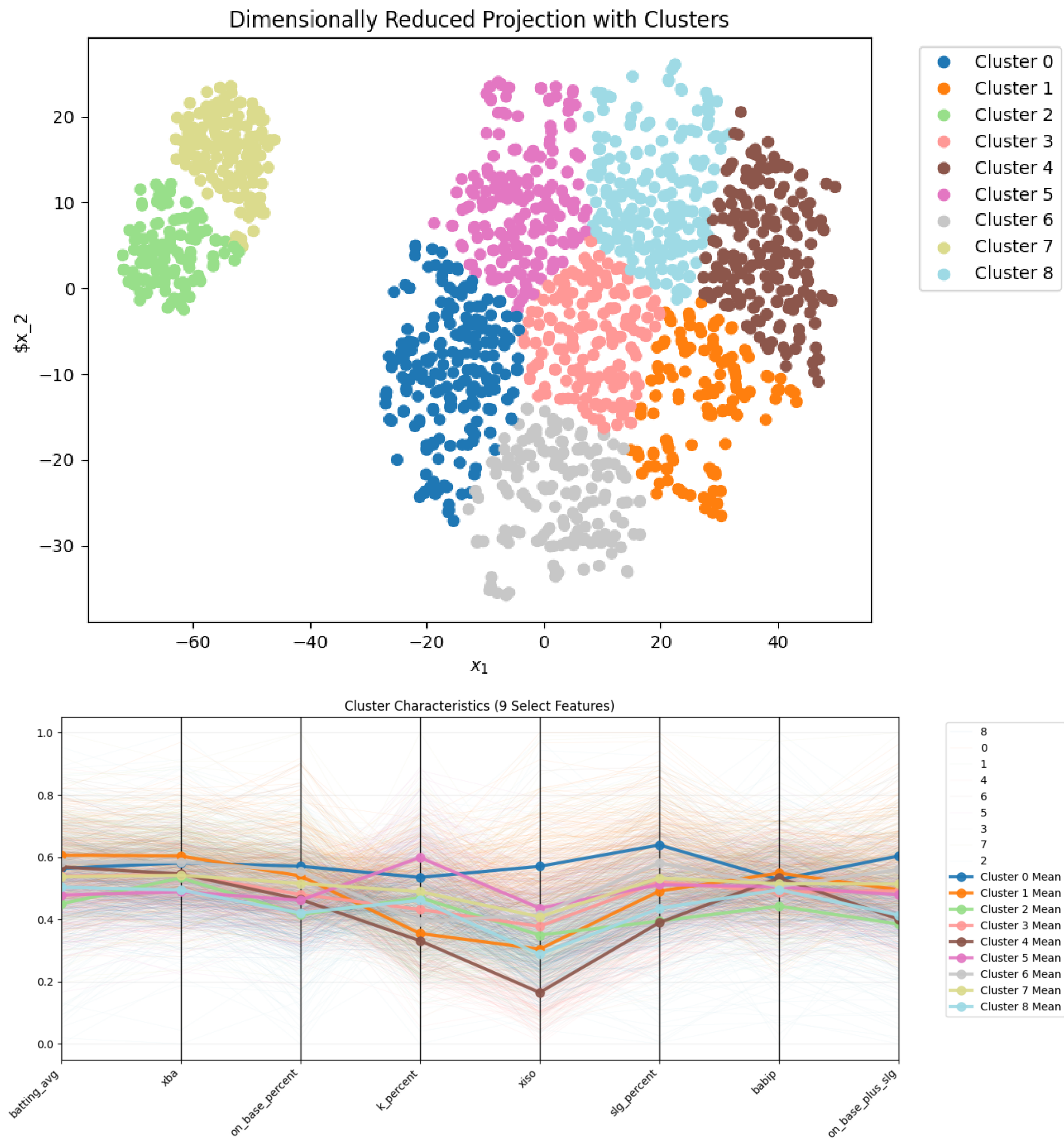

Figure 6: Parallel plot for a cluster resulting from using t-SNE reduction and K-Means clustering. The features chosen are the same features from Table 1 for batters.

	S-Score (# of Clusters)			DB Score (# of Clusters)			CH Score (# of Clusters)		
	PCA	t-SNE	MDS	PCA	t-SNE	MDS	PCA	t-SNE	MDS
Affinity	0.227(27)	0.313(34)	0.255(37)	0.558(37)	0.883(37)	0.687(31)	497.966(37)	2226.112(35)	642.452(37)
Agglomerative	0.689(2)	0.610(2)	0.534(2)	0.456(2)	0.436(2)	0.663(2)	3253.286(2)	2494.279(6)	1618.138(3)
Birch	0.689(2)	0.610(2)	0.534(2)	0.456(2)	0.436(2)	0.663(2)	3253.286(2)	2687.683(4)	1557.104(3)
DBSCAN	0.689(2)	0.610(2)	-1(1)	0.456(2)	0.436(2)	∞(1)	3253.286(2)	2248.510(2)	-1(1)
Gaussian	0.689(2)	0.610(2)	0.534(9)	0.456(2)	0.436(2)	0.663(2)	3253.286(2)	2933.302(5)	1633.854(3)
k-Means	0.689(2)	0.610(2)	0.534(2)	0.456(2)	0.438(2)	0.663(2)	3253.286(2)	2980.003(5)	1635.646(3)
MeanShift	0.689(2)	0.610(2)	0.534(2)	0.456(2)	0.436(2)	0.663(2)	3253.286(2)	2623.886(5)	1416.479(2)
Spectral	0.689(2)	0.610(2)	0.534(2)	0.456(2)	0.436(2)	0.663(2)	3253.286(2)	2773.019(3)	1431.594(8)

Table 3: The three different scores based on dim. reduction and clustering technique for batters

However, one of the purposes of this project was to try to find different ways to see how players perform, not necessarily to see if they perform well or poorly. With that in mind, it is better to look at the different techniques and see which ones generate many clusters but have good scores. This is a problem in the Batters dataset since the best-scoring clusters often have a low (less than four) number of clusters. If a couple of techniques were chosen, it would probably be Gaussian, K-Means, and Mean Shift, since they have the highest CH scores while generating five clusters. For the pitchers, there are more clusters on average, making it better to serve the project's purpose. Again, if a couple of techniques were to be chosen, then the Gaussian and K-Means techniques would be selected for the number of clusters they generate while having a high CH score.

The best CH scoring cluster for K-Means was used for the parallel plotting to analyze different important features across clusters. The parallel plots undoubtedly give more information as to what type of player would be in each cluster and also if that player would be "good" or "bad." For instance, for the pitchers clusters, Cluster 2 consistently is either worse or mediocre than the other clusters in terms of its features being high or low, especially with it having the worst amount of groundouts. Therefore, it would be safe to assume that bad players would go in Cluster 2. However, there are some other informative features in the parallel plots. For instance, Cluster 0 doesn't have the best features out of all the clusters, but they do have a high amount of groundouts, which could imply that pitchers who are in this cluster may not be the best at pitching, but are great at pitching to make groundouts. This could lead further to discovering the player types that might be particular clusters. Switching datasets, the other parallel plot is also informative. For example, Cluster 0 has one of the highest metrics for success from hitting the ball, but also has the second highest strikeout percentage. This could mean that players in Cluster 0 are often looking for hits, even if they potentially could strike out. Overall, the parallel plots successfully show feature importance per cluster.

5.2 Potential Future Changes

One aspect of the project that should be investigated is how much of the data helps determine a player's performance. For instance, the pitcher dataset contains data about different balls the pitchers throw, the percentage of the total pitches of each type, and various information about each pitch. It is unclear whether such extensive data like that would benefit the clustering in finding different player types. It is certainly worth examining whether or not that data should be removed.

Another aspect of the project that could be analyzed is the order of reduction and clustering. In this project, the data was reduced first and then clustered. It might be worth investigating if clustering the data first and then decreasing it while trying to keep the same clusters yields better and/or more informative results. Additionally, more work could be done to find the optimal hyperparameters involved in reducing the data and clustering it. The methods used in this project for determining the hyperparameters were pseudorandom. Still, there might be some use in trying to find better hyperparameters (especially with the Affinity Propagation, since no other hyperparameters other than the default could be used).

6 Conclusion

This project used different data reduction, clustering, and scoring methods to evaluate player types and performance based on hundreds of features. Eight different clustering algorithms were used, along with three different data reduction algorithms and scoring methods, to try to determine the best cluster model. The results of these models showed that while fewer clusters resulted in better scores overall, the lack of variety in the clusters resulted in a lack of variety in the features. The parallel plots were used with multiple clusters to show that each cluster has a different combination of features that defines the players in the cluster. Future work on this project would work on hyperparameter tuning, unimportant feature removal, and reduction-clustering order to see if the resulting clusters are better than what is presented in this paper.

Appendix

The relevant code with plots is attached to the zip file provided with the paper.